Topics Covered:
Physics: Motion in a Plane, Laws of Motion
Chemistry: Structure of Atom, Classification of Elements and Periodicity in Properties
Botany: Cell Cycle & Cell Division, The Living World, Biological Classification-I: (Upto thermoacidophiles)
Zoology: Biomolecules-II: (Proteins, types & functions, Lipids, Nucleic acids, Enzymes, Cofactors), Breathing & Exchange of Gases-I: (Upto mechanism of breathing)

General Instructions :
Duration of Test is 3 hrs.
The Test consists of 180 questions. The maximum marks are 720.
There are four parts in the question paper consisting of Physics, Chemistry, Botany and Zoology having 45 questions in each part of equal weightage.
Each question carries +4 marks. For every wrong response, –1 mark shall be deducted from the total score. Unanswered/unattempted questions will be given no marks.
Use blue/black ballpoint pen only to darken the appropriate circle.
Mark should be dark and completely fill the circle.
Dark only one circle for each entry.
Dark the circle in the space provided only.
Rough work must not be done on the Answer sheet and do not use white fluid or any other rubbing material on the Answer sheet.

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PHYSICS

1. A vector is not changed if
   (1) It is rotated through an arbitrary angle
   (2) It is multiplied by an arbitrary scalar
   (3) A unit vector is added to the vector itself
   (4) It slides parallel to itself

2. If angle between two vectors $\vec{a}$ and $\vec{b}$ is $\frac{\pi}{6}$, then angle between $\vec{a}$ and $-2\vec{b}$ will be
   (1) $\frac{\pi}{3}$
   (2) $\frac{\pi}{6}$
   (3) $\frac{2\pi}{3}$
   (4) $\frac{5\pi}{6}$

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3. Given below are two statements:
   Statement I: A vector quantity has both, magnitude and direction and it must follow triangle law of addition.
   Statement II: If we multiply a vector $\vec{A}$ with a number λ, then $\left|\lambda\vec{A}\right| = \lambda\left|\vec{A}\right|$.
   In the light of the above statements, choose the most appropriate answer from the options given below.
   (1) Statement I is correct but statement II is incorrect
   (2) Both statement I and statement II are correct
   (3) Both statement I and statement II are incorrect
   (4) Statement I is incorrect but statement II is correct

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4. The angle that the vector $\vec{OA} = 5\hat{i} + 5\hat{j}$ makes with y-axis is
   (1) 30°
   (2) 45°
   (3) 60°
   (4) $\tan^{-1}\left(\frac{1}{2}\right)$

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5. If a vector p is making angles α, β and γ respectively with the x, y and z axes, then $\cos^2\alpha + \cos^2\beta + \cos^2\gamma$ is equal to

(1) 0

(2) 1

(3) 2

(4) 3

6. A particle has initial velocity $(6\hat{i} + 8\hat{j})$ m/s and a constant acceleration $\vec{a} = (\hat{i} - \hat{j})$ m/s^2. The velocity of the particle after 2 seconds will be

(1) $(6\hat{i} + 8\hat{j})$ m/s

(2) $(2\hat{i} - 4\hat{j})$ m/s

(3) $(3\hat{i} + 9\hat{j})$ m/s

(4) $(3\hat{i} + \hat{j})$ m/s

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7. A body moves in $(x - y)$ plane, so that the displacement along the x and y axis are given by $x = 2t^3$ m and $y = 4t^2$ m. The speed of body at $t = 1$ s is

(1) 14 m/s

(2) 10 m/s

(3) 24 m/s

(4) 12 m/s

8. Path of projectile with respect to another projectile (when both are in air) is

(1) Circular

(2) Straight line

(3) Hyperbolic

(4) Parabolic

9. A river is flowing with velocity of 3 m/s. A boat can move with velocity of 4 m/s in still water. The width of river is 200 m. Minimum time required to cross the river is

(1) 40 s

(2) 50 s

(3) 60 s

(4) 70 s

10. At the top of the trajectory of projectile, the

(1) Acceleration is minimum

(2) Velocity is zero

(3) Acceleration is maximum

(4) Acceleration is zero

11. Consider the following statements and choose the correct answer.

Statement (A): In oblique projectile motion under gravity, speed continuously decreases.

Statement (B): In oblique projectile motion under gravity, acceleration remains constant.

(1) Both statement (A) and (B) are correct

(2) Both statement (A) and (B) are incorrect

(3) Statement (A) is incorrect and statement (B) is correct

(4) Statement (A) is correct but statement (B) is incorrect

12. Speed of a projectile at the point of projection is $\sqrt{2}$ times the speed at its maximum height, the angle of projection from horizontal of the projectile is

(1) 30°

(2) 45°

(3) 60°

(4) $\tan^{-1}(\sqrt{2})$

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13. The equation of a projectile is $y = ax - \frac{x^2}{2}$. Its horizontal range is (where x is horizontal and y is vertical distance)

(1) $\frac{2a}{1}$

(2)

(3)

(4)

14. A projectile is projected with initial velocity $(6\hat{i} + 8\hat{j})$ m/s, from the ground. The velocity when it is about to hit the ground is (assuming y-axis is along vertical direction)

(1) $(6\hat{i} + 8\hat{j})$ m/s

(2) $(6\hat{i} - 8\hat{j})$ m/s

(3) $(-6\hat{i} - 8\hat{j})$ m/s

(4) $(-6\hat{i} + 8\hat{j})$ m/s

15. A particle is projected at angle 60° with horizontal with an initial speed $20\sqrt{2}$ m s^{-1}. When it makes an angle 45° with horizontal its speed v is

(1) 10 m s^{-1}

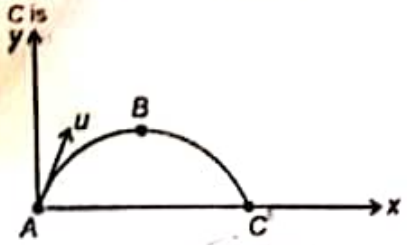
(2) 20 m s^{-1}

(3)

(4)

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16. The velocity of a projectile at the initial point A is $(\hat{i}+\hat{j})$ m/s. Ratio of speed at point B to the speed at point C is



(1) 1

(2) $\sqrt{2}$

(3) $\frac{1}{\sqrt{2}}$

(4) $\frac{1}{2}$

17. A body is projected from ground with velocity $(6\hat{i}+8\hat{j})$ m/s. The time of flight of projectile is (g = 10 m/s^2)

(1) 1.6 s

(2) 1 s

(3) 2 s

(4) 2.6 s

18. A mass is being whirled in the horizontal circle with radius 1 m and angular velocity, $\omega = (t^2 + t)$ rads^{-1}. Net acceleration acting on the mass, at t = 1 s from start is

(1) 5 ms^{-2}

(2) 6 ms^{-2}

(3) 4 ms^{-2}

(4) 3 ms^{-2}

19. A particle is projected with an initial velocity $\vec{v} = 5\hat{i} + 4\hat{j}$ in x-y plane. What will be the equation of trajectory of the particle? (Where x-is horizontal y-is vertical and g = 10 m/s^2)

(1) $y = \frac{5x}{4} - 5x^2$

(2) $y = \frac{4x}{5} - \frac{x^2}{5}$

(3) $y = \frac{4}{5} - \frac{4x^2}{5}$

(4) $y = 5x - \frac{4x^2}{5}$

20. In standard projectile motion, the angle between velocity and acceleration during flight may be

(1) 90°

(2) Acute

(3) Obtuse

(4) All of these

21. A cricket ball is thrown by a player at a speed of 20 m/s in a direction 30° above the horizontal. The maximum height attained by the ball during its motion is (g = 10 m/s^2)

(1) 25 m

(2) 5 m

(3) 10 m

(4) 20 m

22. A particle moves in x-y plane according to the law $\vec{v} = 3y^2\hat{i} + 2x\hat{j}$. The equation of trajectory is

(1) $3y^2 = 2x$

(2) $\frac{x^2}{2} - \frac{y^3}{6}$ = constant

(3) $\frac{x^2}{4} - \frac{y^3}{3}$ = constant

(4) $x^2 - y^3$ = constant

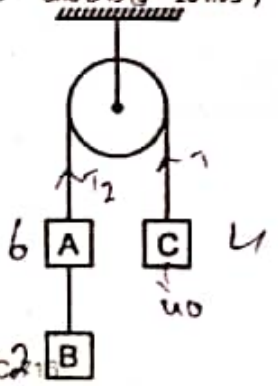
23. A force of $\vec{F} = (2\hat{i} - 3\hat{j})$ N produces an acceleration of $2\sqrt{2}$ m/s^2 on a body. The mass of the body is

(1) $\sqrt{8}$ kg

(2) $\sqrt{12}$ kg

(3) [illegible]

24. Three blocks A, B and C of masses 6 kg, 2 kg and 4 kg respectively, are connected together with the help of light inextensible strings passing over a frictionless pulley as shown in figure. Then the tension in the string connecting A and B is (g = 10 m/s^2)



(1) $\frac{20}{3}$ N

(2) $\frac{40}{3}$ N

(3) $\frac{80}{3}$ N

(4) $\frac{160}{3}$ N

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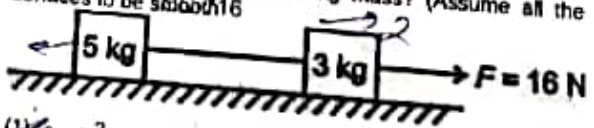
25. A force acts for 8 s on a body of mass 5 kg at rest kept on a smooth horizontal surface. The force then stops acting and the body describes 80 m in next 5 s. The force acting on body is

(1) 10 N
(2) 15 N
(3) 20 N
(4) 25 N

26. If force-time relation for a body is given by $F = (2t^2 + 4t + 1)$ N, then the change in momentum of the body from $t = 0$ to $t = 1$ s will be

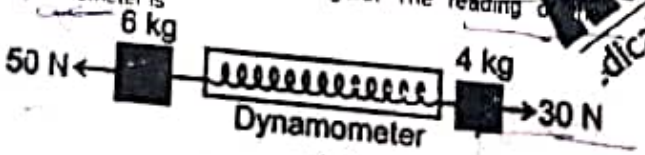
(1) $\frac{11}{3}$ kg m/s
(2) $\frac{11}{2}$ kg m/s
(3) $\frac{8}{3}$ kg m/s
(4) $\frac{7}{3}$ kg m/s

27. What is the acceleration of 5 kg mass? (Assume all the surfaces to be smooth)



(1) 2 m/s^2
(2) 4 m/s^2
(3) 1 m/s^2
(4) 3 m/s^2

28. A dynamometer D is attached to two blocks of mass 6 kg and 4 kg as shown in the figure. The reading of the dynamometer is

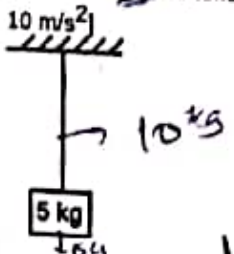


(1) 18 N
(2) 28 N
(3) 38 N
(4) 48 N

29. A bullet is fired with a speed 200 m/s. If the mass of bullet is 100 g and that of gun is 2 kg, then the recoil speed of gun is

(1) 20 m/s
(2) 2 m/s
(3) 10 m/s
(4) 5 m/s

30. In the shown figure, mass of the rope is 10 kg and of the block is 5 kg. The tension in the rope at topmost point is (g = 10 m/s^2)



(1) 150 N
(2) 50 N
(3) 100 N
(4) 75 N

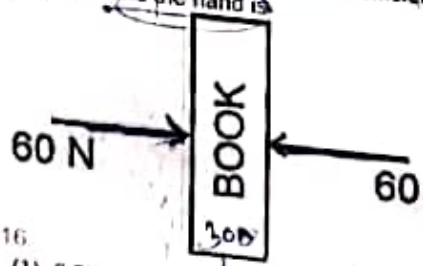
31. A block of mass 2 kg lies on a horizontal surface in a truck. The coefficient of friction between the block and surface is 0.5. If the acceleration of the truck is 4 m/s^2, then friction force acting on the block will be (g = 10 m/s^2)

(1) 10 N
(2) 8 N
(3) 6 N
(4) Zero

32. The coefficient of static friction between a block and an inclined plane is $\frac{1}{\sqrt{3}}$, then the angle of repose for the inclined plane is

(1) 60°
(2) 45°
(3) 30°
(4) 90°

33. A book of weight 30 N is pressed between two hands and each hand exerts a force of 60 N as shown. If the book is just about to slide down, then coefficient of friction between the book and the hand is



(1) 0.25
(2) 0.3
(3) 0.5
(4) 0.4

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34. A block of mass 2 kg is resting on a rough horizontal surface with a coefficient of kinetic friction $\mu_k = 0.6$. If a horizontal force of 28 N is applied to the block, find its velocity after 1 s of motion starting from rest. ($g = 10\ m/s^2$)

(1) 8 m/s

(2) 7 m/s

(3) 10 m/s

(4) 5 m/s

35. Consider the following statements
Statement-A: If a body is acted upon by a single external force, it cannot be in equilibrium.
Statement-B: If a body is in equilibrium under the action of only two external forces, the forces must be equal and opposite.
Statement-C: If a body is in equilibrium under the action of three forces, then they may not be coplanar.
Choose the correct option.

(1) Only statement A is correct

(2) All A, B and C are correct

(3) Both A and C are correct

(4) Both A and B are correct

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36. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R)
Assertion (A): Pseudo force can only be felt in a non-inertial frame.
Reason (R): A frame in uniform motion is a non-inertial frame.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) (A) is false but (R) is true

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37. A boat moves relative to water with a velocity $\vec{v}_{bw} = 2\hat{j}$ m/s across a river flowing with a velocity $\vec{v}_w = 2\hat{i}$ m/s. The boat covers a displacement of $\sqrt{2}$ m relative to the ground in a time $t = \frac{\pi}{4}$ s. Match the physical quantities given in List-I with their corresponding values in List-II:

List-I	List-II
(A) Speed of the boat with respect to water	(i) Zero
(B) Magnitude of average velocity of boat relative to ground over time $t = \frac{\pi}{4}$ s	(ii) $2\sqrt{2}$ m/s
(C) Magnitude of velocity of the boat with respect to ground	(iii) 2 m/s
(D) Component of velocity of the boat relative to water along the flow of river	(iv) $\frac{4\sqrt{2}}{\pi}$ m/s

Choose the correct answer from the options given below

(1) (A)-(iii), (B)-(iv), (C)-(ii), (D)-(i)

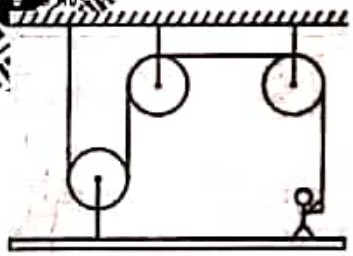
(2) (A)-(iii), (B)-(iv), (C)-(i), (D)-(ii)

(3) (A)-(iv), (B)-(iii), (C)-(ii), (D)-(i)

(4) (A)-(iv), (B)-(iii), (C)-(i), (D)-(ii)

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38. A 50 kg person stands on a 25 kg platform. He pulls on the rope which is attached to the platform via the frictionless pulley system shown in the figure. The whole system remains in equilibrium. The force applied by the person on the rope is [g = 10 m/s^2]



(1) 30 N

(2) 60 N

(3) 120 N

(4) 250 N

39. A monkey is descending from the branch of a tree with a constant acceleration. If the breaking strength of the branch is 75% of the weight of the monkey, the minimum acceleration with which the monkey can climb down without breaking the branch is

(1) g

(2) $\frac{g}{2}$

(3) $\frac{g}{4}$

(4) $\frac{3g}{4}$

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40. A projectile is launched from horizontal ground with an initial speed $u = 20$ m/s at an angle θ to the horizontal. If the maximum height reached by the projectile is 5 m, find the angle of projection θ. ($g = 10$ m/s^2)

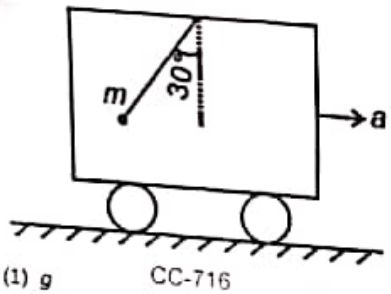
(1) 15°

(2) 30°

(3) 45°

(4) 60°

41. If trolley accelerates horizontally with acceleration a, then bob is displaced backward from its initial vertical position by an angle 30° as shown in figure. The value of a in terms of g is



(1) g

(2) $\sqrt{3}g$

(3) $\frac{g}{\sqrt{3}}$

(4) $\frac{g}{2}$

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42. A batsman hits back a ball straight in the direction of the bowler without changing its initial speed of 12 ms^{-1}. If the mass of the ball is 0.15 kg the impulse imparted to the ball is

(1) 1.5 Ns

(2) 3.6 Ns

(3) 7.2 Ns

(4) 10.2 Ns

43. If a body is moving with constant acceleration, then

(1) Only a single force is acting on it

(2) Multiple forces may act on the body

(3) Net force acting on the body are variable

(4) Net force acting on the body may be zero

44. The momentum of a particle varies with time as $P = 2t + 3t^2$ (where P is in kg m/s and t is in seconds). The force acting on particle at $t = 2$ s is

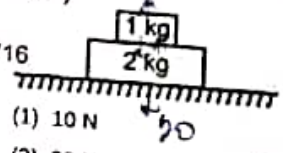
(1) 14 N

(2) 12 N

(3) 16 N

(4) 20 N

45. A block of mass 1 kg is placed on the another block of mass 2 kg which is placed on ground as shown in figure. The normal force on the 2 kg block by the ground will be ($g = 10$ m/s^2)



(1) 10 N

(2) 20 N

(3) 30

(4)

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46. The wavenumber of radiation having wavelength 5000 Å is

(1) 2×10^{-7} cm^{-1}

(2) 2×10^{-5} cm^{-1}

(3) 2×10^{4} cm^{-1}

(4) 2×10^{5} cm^{-1}

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47. The formula of halide of a metal whose successive ionization enthalpies are x, 2x, 5x, 100x kJ mol^{-1} respectively is

(1) MX

(2) MX_2

(3) MX_3

(4) M_2X

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48. Consider the following statements about photoelectric effect
(a) There is no time lag between the striking of light beam and the ejection of electrons from metal surface.
(b) The number of electrons ejected is directly proportional to the frequency of light.
(c) The kinetic energy of ejected electrons increases with the increase of intensity of light used.
The correct statement(s) is/are

(1) (a) only

(2) (a) and (b) only

(3) (b) and (c) only

(4) (a), (b) and (c)

49. The ratio of wavelength λ_2 to λ_1 for two radiations is 1:2. The possible value of energies E_1 and E_2 respectively can be

(1) 25 and 75

(2) 25 and 50

(3) 50 and 25

(4) 50 and 75

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50. A 100 watt bulb emits monochromatic light of wavelength 400 nm. The number of photons emitted per second by the bulb is

(1) 1×10^{15}

(2) 2×10^{20}

(3) 2×10^{10}

(4) 8×10^{20}

51. Energy of an electron (in eV/atom) of 4th Bohr orbit of Li^{2+} ion is

(1) –13.6

(2) –27.2

(3) –3.4

(4) –6.8

52. Correct order of penetrating power of α-particles, β-rays and γ-rays is

(1) γ-rays > β-rays > α-particles

(2) α-particles > β-rays > γ-rays

(3) γ-rays > α-particles > β-rays

(4) β-rays > α-particles > γ-rays

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53. Which of the following is an endothermic process?

(1) $F(g) + e^- \rightarrow F^-(g)$

(2) $S(g) + e^- \rightarrow S^-(g)$

(3) $O^-(g) + e^- \rightarrow O^{2-}(g)$

(4) $O(g) + e^- \rightarrow O^-(g)$

54. The correct set of quantum numbers for the valence shell electron of Ca (Atomic number : 20) is

(1) $n = 4, l = 0, m = 0, s = -\frac{1}{2}$

(2) $n = 3, l = 0, m = 0, s = +\frac{1}{2}$

(3) $n = 3, l = 1, m = 0, s = +\frac{1}{2}$

(4) $n = 3, l = 0, m = 0, s = -\frac{1}{2}$

55. If shortest wavelength in Lyman series of H-atom be 'x' Å, then the longest wavelength in Balmer series of He^+ would be (in Å)

(1) $\frac{36x}{5}$

(2) x

(3) $\frac{9x}{5}$

(4) $\frac{27x}{20}$

56. The correct order of energy of given orbitals in multielectron species is

(1) $4d > 5p > 5s$

(2) $5p > 5s > 4d$

(3) $5p > 4d > 5s$

(4) $5s > 5p > 4d$

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57. Si

(1) B

(2) Ga

(3) Si

(4) Ge

58. Maximum number of electrons that can be associated with the set of quantum numbers n = 2, l = 1 is

(1) 2

(2) 3

(3) 5

(4) 6

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59. Match List-I with List-II.

	List I (Electronic transition)		List II (Maximum number of emission lines in H atom sample)
a.	$n_2 = 5$ to $n_1 = 1$	(i)	3
b.	$n_2 = 3$ to $n_1 = 2$	(ii)	10
c.	$n_2 = 4$ to $n_1 = 2$	(iii)	1
d.	$n_2 = 6$ to $n_1 = 1$	(iv)	15

Choose the correct option.

(1) a(i), b(ii), c(iii), d(iv)

(2) a(iv), b(iii), c(ii), d(i)

(3) a(iii), b(i), c(iv), d(ii)

(4) a(ii), b(iii), c(i), d(iv)

60. Total number of elements present in 3rd period of modern periodic table is

(1) 18

(2) 32

(3) 36

(4) 8

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61. The first ionisation potential of Na is 5.1 eV. The value of electron gain enthalpy of Na^+ will be

(1) −5.1 eV

(2) −10.2 eV

(3) +2.55 eV

(4) −2.55 eV

62. When an electron falls from 2nd energy level to ground state then which of the following will produce shortest wavelength?

(1) Be^{3+} ion

(2) D-atom

(3) H-atom

(4) He^+ ion

63. Orbital angular momentum of electron present in a 3p orbital is

(1) $\sqrt{12}\hbar$

(2) $\sqrt{6}\hbar$

(3) $\sqrt{2}\hbar$

(4) $4\hbar$

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64. According to the Bohr's theory, which of the following transitions in the hydrogen atom will give rise to the most energetic photon?

(1) n = 5 to n = 3

(2) n = 4 to n = 3

(3) n = 3 to n = 1

(4) n = 6 to n = 5

65. Number of electrons in nitrogen atom for which azimuthal quantum number(l) is 1, is

(1) 7

(2) 3

(3) 4

(4) 6

66. Angular momentum of an electron in the third orbit of hydrogen atom is

(1) $\frac{3h}{2\pi}$

(2) $\frac{h}{2\pi}$

(3) $\frac{3h}{4\pi}$

(4) $3h$

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67. Match the species given in List-I with their respective isoelectronic species given in List-II

List-I

a. O

b. NO

c. CO^+

d. CO

List-II

(i) F_2

(ii) N_2^+

(iii) C_2^{2-}

(iv) N_2^-

Choose the correct option.

(1) a(iii), b(i), c(iv), d(ii)

(2) a(i), b(iii), c(ii), d(iv)

(3) a(iii), b(i), c(ii), d(iv)

(4) a(i), b(iii), c(iv), d(ii)

68. Energy of 2s orbital of which element is lowest?

(1) K

(2) H

(3) Li

(4) Na

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69. Consider the following pairs of elements.
(a) O and Cl
(b) S and Br
(c) Li and Mg
(d) Al and Be
The pairs which show diagonal relationship are
(1) (a), (b) and (c) only
(2) (b) and (c) only
(3) (b) and (d) only
(4) (c) and (d) only

70. In the following configuration
$ns^2(n-1)d^{0-1}(n-2)f^{1-14}$
If value of (n –1) = 5, then the configuration will be of
(1) Actinoid
(2) Representative element
(3) Transition element
(4) Lanthanoid

71. Match the following items in column I with column II and choose the correct option.

	Column I (quantum no. s)		Column II (orbital)
(a)	n = 2, l = 1	(i)	3p
(b)	n = 3, l = 2	(ii)	3d
(c)	n = 1, l = 0	(iii)	2p
(d)	n = 3, l = 1	(iv)	1s

(1) (a)(iii), (b)(i), (c)(iv), (d)(ii)
(2) (a)(iii), (b)(ii), (c)(iv), (d)(i)
(3) (a)(ii), (b)(i), (c)(iii), (d)(iv)
(4) (a)(i), (b)(ii), (c)(iii), (d)(iv)

72. Which among the following electronic configurations belong to main group elements?
A. [Ne]$3s^1$
B. [Ar]$3d^3 4s^2$
C. [Kr]$4d^{10} 5s^2 5p^5$
D. [Ar]$3d^{10} 4s^1$
E. [Rn]$5f^0 6d^2 7s^2$
Choose the correct answer from the option given below:
(1) D and E only
(2) A, C and D only
(3) B and E only
(4) A and C only

73. Which of the following sets of options is not a Dobereiner's Triad?
(1) Li, Na, K
(2) Be, Mg, Ca
(3) Ca, Sr, Ba
(4) Cl, Br, I

74. The correct increasing order of atomic radii for the following species is
(1) Be < Li < Mg < Na
(2) Li < Be < Mg < Na
(3) Li < Be < Na < Mg
(4) Be < Li < Na < Mg

75. Which of the following has the highest negative electron gain enthalpy?
(1) Bromine
(2) Oxygen
(3) Sulphur
(4) Chlorine

76. Which of the following oxides is neutral?
(1) CaO
(2) N_2O
(3) SO_3
(4) Cl_2O_7

77. The energy of one mole of photons of radiation whose frequency is 3×10^{14} Hz is
(1) 300 kJ
(2) 150 kJ
(3) 150.7 kJ
(4) 299.6 kJ

78. The most electronegative element has the following electronic configuration
(1) ns^2np^1
(2) ns^2np^3
(3) ns^2np^5
(4) ns^2np^6

79. Match List-I with List-II.

List-I (Symbol)	List-II (IUPAC Official Name)
(a) Unb	(i) Mendelevium
(b) Unq	(ii) Seaborgium
(c) Unh	(iii) Rutherfordium
(d) Unu	(iv) Nobelium

Choose the correct match.
(1) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)
(2) (a)-(iv), (b)-(iii), (c)-(i), (d)-(ii)
(3) (a)-(i), (b)-(ii), (c)-(iv), (d)-(iii)
(4) (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i)

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80. Choose the incorrect statement about canal rays.

(1) They possess positive charge

(2) They travel in a straight line in the absence of electrical or magnetic field

(3) They emerge from anode and move towards the cathode

(4) Properties of canal rays depend upon the nature of the gas present in cathode ray tube

81. Match the following columns I and II.

	Column I (Element)		Column II (Number of unpaired electron(s))
a.	Copper	(i)	5
b.	Manganese	(ii)	4
c.	Chromium	(iii)	1
d.	Iron	(iv)	6

Choose the correct option.

(1) a(ii), b(iv), c(i), d(iii)

(2) a(ii), b(iii), c(iv), d(i)

(3) a(iii), b(i), c(ii), d(iv)

(4) a(iii), b(i), c(iv), d(ii)

82. If a particle of mass 0.01 g is moving with a velocity of 200 m/s then the de-Broglie wavelength associated with the moving particle will be ($h = 6.6 \times 10^{-34}$ Js)

(1) 3.3×10^{-34} m

(2) 3.3×10^{-30} m

(3) 3.3×10^{-32} m

(4) 3.3×10^{-31} m

83. Match List I with List II and choose the correct option.

List I (Metals)	List II (Work function/eV)
(a) Li	(i) 2.25
(b) Na	(ii) 2.42
(c) K	(iii) 3.7
(d) Mg	(iv) 2.3

(1) (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv)

(2) (a)-(i), (b)-(iv), (c)-(iii), (d)-(ii)

(3) (a)-(ii), (b)-(iv), (c)-(i), (d)-(iii)

(4) (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)

84. The threshold frequency for a metal is $8 \times 10^{14} s^{-1}$. What is the K.E. of an electron emitted when radiation of frequency $2 \times 10^{15} s^{-1}$ hits the metal

(1) 7.5×10^{-20} J

(2) 7.25×10^{-19} J

(3) 7.0×10^{-15} J

(4) 7.95×10^{-19} J

85. The de Broglie wavelength of the electron in the first orbit of Li^{2+} ion is (Given Bohr radius, $a_0 = 52.9$ pm)

(1) 105.8π pm

(2) 52.9π pm

(3) 35.3π pm

(4) 158.7π pm

86. Series of hydrogen spectrum which lies in ultraviolet region is

(1) Lyman

(2) Balmer

(3) Paschen

(4) [illegible]

87. Energy required for ionisation of 0.04 gram atom of magnesium is x kJ. The amount of energy required to ionise 20 atoms of magnesium is (N_A = Avogadro's number)

(1) 500 x kJ

(2) $\frac{250x}{N_A}$ kJ

(3) $\frac{25\times10^{3}x}{N_A}$ kJ

(4) $\frac{N_A}{25\times10^{3}x}$ kJ

88. Orbital in which electron density is symmetrically distributed is

(1) 1s

(2) $2p_x$

(3) $3p_y$

(4) $4d_{xy}$

89. The element A (atomic number = 39) is present in which group of periodic table?

(1) First

(2) Third

(3) Fifteenth

(4) Tenth

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90. Given below are two statements.

Statement I : The radius of 2nd orbit of He^+ ion is 1.058 Å.
Statement II : Splitting of spectral line in electrical field is called Stark effect.
In light of above statements, choose the **correct** answer.

(1) Statement I is correct but statement II is incorrect

(2) Statement I is incorrect but statement II is correct

(3) Both statement I and statement II are incorrect

(4) Both statement I and statement II are correct

BOTANY

91. Read the given statements and select the **correct** option.
Statement A: Cell growth results in disturbing the ratio between the nucleus and the cytoplasm.
Statement B: Mitotic divisions in the meristematic tissues-the apical and the lateral cambium, result in a continuous growth of plants throughout their life.

(1) Only statement A is correct

(2) Only statement B is correct

(3) Both the statements A and B are correct

(4) Both the statements A and B are incorrect

92. Select the **correct** sequence of events which takes place during meiosis II.
a. Condensation of chromosome
b. Formation of four haploid cells
c. Chromosome align at the equator

(1) a → b → c

(2) b → c → a

(3) a → c → b

(4) b → a → c

93. Read the following statements and choose the **correct** option.
Statement A: A human cell in culture divides approximately in every 24 minutes.
Statement B: Yeast can progress through the cell cycle in only about 90 minutes.

(1) Only statemnt A is incorrect

(2) Only statement B is incorrect

(3) Both statements A and B are correct

(4) Both statements A and B are incorrect

94. During cytokinesis in an animal cell

(1) Furrow formation starts at the centre of the cell and grows outwards

(2) Furrow formation starts at the periphery and then moves inwards centripetally

(3) Phragmoplast is formed by golgi complex

(4) Cytoplasm divides centrifugally

95. Quiescent stage of cell cycle shows

(1) Metabolic inactivity

(2) High rate of proliferation

(3) Division only when they are called to do so

(4) Condensation of chromatin material

96. Which one of the following precedes re-formation of the nuclear envelope during M phase of the cell cycle?

(1) Distribution of plastids among daughter cells and the formation of phragmoplast

(2) Exchange of genetic material between two homologous chromosomes and formation of middle lamella

(3) Chromosomes cluster at opposite spindle poles and their decondensation

(4) Formation of endoplasmic reticulum and chromosome duplication in interkinesis

97. The two daughter cells formed during mitosis contain

(1) The same amount of DNA but a set of chromosomes different from those of parental cell

(2) The same amount of DNA and the same set of chromosomes as those of the parent cell

(3) Half the amount of DNA and the same set of chromosomes as those of the parent cell

(4) Double the amount of DNA and a set of chromosomes different from those of the parent cell

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98. Read the following Assertion(A) and Reason (R) statements and choose the correct option

A : In telophase stage of mitosis two daughter nuclei are formed.

R : In telophase stage, nucleolus, endoplasmic reticulum, golgi bodies and other organelles reappear in the daughter cells.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion.

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion.

(3) Assertion is true statement but Reason is false.

(4) Both Assertion and Reason are false statements.

99. "A cell has 15 bivalents in its zygotene stage". Choose the correct statement associated with this cell.

(1) The cell has total 60 chromosomes

(2) Each of the four daughter cells produced from the above cell will have 15 chromosomes

(3) This cell is undergoing mitosis

(4) This cell has total 60 chromatids.

100. If karyokinesis is not followed by cytokinesis, then it gives rise to

(1) Zygote

(2) Fertilised egg

(3) Multinucleate condition

(4) Embryo

101. How many meiosis are required to form 100 pollen grains from microspore mother cells?

(1) 100

(2) 50

(3) 25

(4) 200

102. Diplotene stage is recognised by

(1) Formation of synaptonemal complex

(2) Dissolution of synaptonemal complex

(3) Disappearance of nuclear membrane and nucleolus completely

(4) Terminalisation of chiasmata

103. Match the following

	Column I		Column II
a.	G_1 phase	(i)	Most of the cell organelle duplicates
b.	M phase	(ii)	Centriole duplication will take place
c.	S-phase	(iii)	Quescent phase
d.	G_0 phase	(iv)	Actual cell division takes place

(1) a(ii), b(i), c(iv), d(iii)

(2) a(i), b(iv), c(ii), d(iii)

(3) a(i), b(iv), c(iii), d(ii)

(4) a(iv), b(iii), c(i), d(ii)

104. The microtubules of the spindle fibres attach to the kinetochore during

(1) Anaphase

(2) Prophase

(3) Metaphase

(4) Telophase

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105. _____ marks the end of the cell division.

(1) [illegible]on of metaphasic plate

(2) [illegible] of nucleus into two daughter nuclei

(3) [illegible]ance of Golgi bodies in the cells

(4) [illegible]on of cytoplasm into two daughter cells

106. Best stage to observe acrocentric chromosome in J shape is

(1) Metaphase

(2) Anaphase

(3) Prophase

(4) Telophase

107. In sexually reproducing organisms, conservation of chromosome number of each species is specifically achieved across generation. Which of the following mechanisms play an important role, so that the conservation of number of chromosomes is maintained?

(1) Mitosis

(2) Crossing over

(3) Meiosis

(4) Synapsis

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108. In male honey bees, the cell division is

(1) Meiotic, in which recombination occurs

(2) Usually responsible for disturbing the nucleo-cytoplasmic ratio

(3) Mitosis, resulting in daughter cells with identical genetic complement

(4) Amitosis, as there is no spindle formation

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109. Which of the given is/are **not** observed during cell division in a meristematic cell?

(1) Spindle fibres

(2) Decodensation of chromosomes

(3) Bivalents

(4) Cell plate

110. The phenomena of congression of chromosomes is observed in

(1) Prophase stage

(2) Metaphase stage

(3) Anaphase stage

(4) Telophase stage

111. In meiosis I, the enzyme recombinase participates during

(1) Prophase I

(2) Anaphase I

(3) Metaphase I

(4) Telophase I

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112. During interkinesis stage,

(1) Replication of DNA occurs.

(2) Replication of centrioles occurs in plant cells.

(3) Chromosomes form chromatin fibres.

(4) RNA and proteins are synthesized.

113. Exchange of genetic material between homologous chromosomes occurs during

(1) Replacement of old dead cells

(2) Gametogenesis

(3) Meiosis-II only

(4) Fertilisation

114. Read the following statements and select the **correct** option.

Assertion (A): Archaebacteria can survive in extreme conditions.

Reason (R): Archaebacteria have a different cell wall structure from other bacteria.

(1) (A) is true and (R) is false

(2) (A) is false and (R) is true

(3) Both (A) and (R) are true and (R) is correct explanation of (A)

(4) Both (A) and (R) are true but (R) is not correct explanation of (A)

115. Match the following columns and select the **correct** option.

	Column-I		Column-II
a.	Taxonomy	(i)	These are distinct biological entities and not merely morphological aggregates
b.	Systematics	(ii)	A process of classification, based on characteristics, all living organisms can be classified into different taxa
c.	Classification	(iii)	It takes into account evolutionary relationships between organisms
d.	Taxonomic categories	(iv)	Process by which anything is grouped into convenient categories based on some easily observable characters

(1) a(iii), b(iv), c(i), d(ii)

(2) a(ii), b(iii), c(iv), d(i)

(3) a(ii), b(i), c(iv), d(iii)

(4) a(iv), b(iii), c(i), d(ii)

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116. Select the **incorrect** match w.r.t. housefly.

(1)	*Musca*	–	Species
(2)	Insecta	–	Class
(3)	Muscidae	–	Family
(4)	Arthropoda	–	Phylum

(1) (2) (3) (4)

117. Read the following statements and choose the correct option.

Statement-A : Modern taxonomic studies are based on external and internal structure of organisms along with cell structure, development process and ecological information of organisms.

Statement-B : The classical taxonomy is based on observable morphological characters.

(1) Only statement-A is incorrect

(2) Only statement-B is incorrect

(3) Both statements are incorrect

(4) Both statements are correct

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118. Potato and mango belong to the same

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(1) Family

(2) Order

(3) Genus

(4) Class

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119. Which of the following is not true?

(1) Genera are aggregates of closely related species.

(2) Plant family Convolvulaceae is included in the order Polymoniales, mainly on the basis of vegetative characters.

(3) The organisms with dorsal hollow neural system belong to the phylum, Chordata.

(4) Lower the taxa, more are the characteristics that the members within the taxon share.

120. Choose the scientific name that is printed **correctly**, following the universal rules of Binomial nomenclature system.

(1) Mangifera *indica* linn.

(2) *Panthera* leo *L*

(3) *Homo* Sapiens Linn

(4) *Triticum aestivum* Linn.

121. Select the statement that is **odd one** w.r.t. binomial nomenclature of organisms.

(1) This naming system was given by Carolus Linnaeus

(2) The first word in this system represents the specific epithet

(3) Names are generally in Latin or latinised irrespective of their origin

(4) Names are printed in italics to indicate their Latin origin

122. "Tiger is a mammal and mammals are animals". In the given statement, "Tiger, Mammals and Animals" represent

(1) Taxa of same category

(2) Taxa of different levels

(3) Similar taxa at same level

(4) Genus, class and phylum respectively

123. Select the option showing hierarchial arrangement of taxonomic categories in ascending order

(1) Species → Genus → Family → Order

(2) Genus → Species → Family → Order

(3) Genus → Species → Order → Family

(4) Species → Family → Order → Genus

124. Mango belongs to the order

(1) Dicotyledonae

(2) Monocotyledonae

(3) Poales

(4) Sapindales

125. Select the taxonomic category which is not common between brinjal and potato.

(1) Genus

(2) Species

(3) Family

(4) Order

126. In *Panthera leo*, the term *leo* represents

(1) Specific epithet

(2) The taxa that is an aggregation of closely related species

(3) Group of related families

(4) Abbreviated name of the taxonomist who first described this organism

127. *Datura* is placed in the family

(1) Poaceae

(2) Anacardiaceae

(3) Solanaceae

(4) Muscidae

128. According to IUCN the number of species of plants and animals that are known and described range between

(1) 1.7 to 1.8 billion

(2) 10 to 15 billion

(3) 1.7 to 1.8 million

(4) 10 to 15 billion

129. The main criteria for classification, used by R.H. Whittaker include

(i) Cell structure
(ii) Body organisation
(iii) Mode of nutrition
(iv) Phylogenetic relationship
(v) Reproduction

The **correct** ones are.

(1) (i), (ii) and (iii) only

(2) (i), (iii) and (iv) only

(3) All (i), (ii), (iii), (iv) and (v)

(4) (ii), (iv) and (v) only

130. The bacteria which can live in extreme saline conditions are

(1) Methanogens

(2) Photosynthetic

(3) Halophiles

(4) Mostly found in hot springs

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131. Plants were classified into trees, shrubs and herbs by using simple morphological characters by

(1) Linnaeus

(2) Ernst Mayr

(3) Aristotle

(4) Carl Woese

132. In the classification system proposed by Linnaeus, algae and fungi are kept in the same kingdom. It indicates that this system does not distinguish between

(1) Cellular and acellular organisms

(2) Eukaryotic and prokaryotic organisms

(3) Photosynthetic and non-photosynthetic organisms

(4) Unicellular and multicellular organisms

133. Correct sequence of cell cycle is

(1) $G_1 \rightarrow G_2 \rightarrow S \rightarrow M$

(2) $G_1 \rightarrow S \rightarrow G_2 \rightarrow M$

(3) $M \rightarrow G_2 \rightarrow S \rightarrow G_1$

(4) $S \rightarrow M \rightarrow G_2 \rightarrow G_1$

134. The bacteria found in gut of ruminant animals and produce methane from dung are

(1) Thermoacidophiles

(2) Cyanobacteria

(3) Halophiles

(4) Methanogens

135. The three-domain system of classification

(1) Places Monera in all three domains

(2) Proposes two domains for Eukaryotes

(3) Divides kingdom Monera into two domains

(4) Proposes one domain for all prokaryotes

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ZOOLOGY

136. Conducting part of respiratory system extends from

(1) Trachea upto bronchi

(2) Pharynx upto trachea

(3) External nostrils upto terminal bronchioles

(4) External nostrils upto alveolar ducts

137. 'X' is the prosthetic group of an enzyme named catalase. 'X' is also the prosthetic group in

(1) Carboxypeptidase

(2) Peroxidase

(3) Carbonic anhydrase

(4) Succinic dehydrogenase

138. The R group is a methyl group in amino acid

(1) Glycine

(2) Alanine

(3) Serine

(4) Tryptophan

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139. Which of the following structures of respiratory tract is not supported by incomplete cartilaginous rings?

(1) Trachea

(2) Terminal bronchiole

(3) Primary bronchiole

(4) Secondary bronchiole

140. Volume of air inspired or expired during a normal respiration by a healthy man is approximately

(1) 3000 mL

(2) 2500 mL

(3) 500 mL

(4) 1500 mL

141. Antibodies that help to fight infectious agents are

(1) Polysaccharides

(2) Amino acids

(3) Proteins

(4) Glucose

142. Dietary proteins are the source of ____ which can not be synthesized in our body. Choose the option which fills the blank correctly.

(1) Non-essential amino acids

(2) Essential amino acids

(3) Semi-essential amino acids only

(4) Basic amino acids only

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143. Trachea divides into right and left primary bronchi at which level of thoracic vertebra?

(1) 5th

(2) 7th

(3) 1st

(4) 4th

144. The most abundant protein in whole of the biosphere is

(1) Collagen

(2) RuBisCO

(3) Chitin

(4) Glycogen

145. "All enzymes are proteins". An exception to this is

(1) Ribozyme

(2) Dehydrogenase

(3) Carbonic anhydrase

(4) Trypsin

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146. Match Column-I with Column-II and choose the correct option.

	Column I		Column II
a.	Beta-pleated sheet	(i)	Simplest amino acid
b.	Glycosidic bond	(ii)	Present in tertiary and quaternary structure of proteins
c.	Disulphide bond	(iii)	Exist between two simple sugars in oligosaccharide
d.	Glycine	(iv)	Secondary structure of protein

(1) a(i), b(ii), c(iii), d(iv)

(2) a(iv), b(iii), c(ii), d(i)

(3) a(iv), b(ii), c(iii), d(i)

(4) a(iv), b(i), c(iii), d(ii)

147. Consider the statements given below

Statement A: A triglyceride is composed of one molecule of fatty acid attached to three molecules of glycerol.

Statement B: Lecithin contains three fatty acids in its structure.

Select the correct option.

(1) Both statements A and B are correct

(2) Both statements A and B are incorrect

(3) Only statement A is correct

(4) Only statement B is correct

148. In a DNA molecule, guanine compulsorily base pairs with

(1) Adenine

(2) Thymine

(3) Cytosine

(4) Uracil

149. Expiratory capacity is

(1) TV + ERV

(2) TV + IRV

(3) VC + RV

(4) RV + ERV + TV

150. Which of the following sets of co-enzymes are nucleotides that contain the vitamin niacin?

(1) NAD, NADP

(2) FMN, FAD

(3) ATP, ADP

(4) ATP, FAD

151. At high temperature, enzymes become A, while at low temperature enzymes remain in B state. Choose the option that correctly fills the blanks A and B, respectively

(1) Inoperative, active

(2) Inactivated, active

(3) Denatured, inactive

(4) Inactivated, hyperactive

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152. Select the option that represents an aromatic amino acid.

(1) Alanine

(2) Tyrosine

(3) Arginine

(4) Serine

153. There is a two layered membrane covering the lungs, called

(1) Peritoneum

(2) Pleurae

(3) Pericardium

(4) Duramater

154. Competitive inhibition could be reversed by increasing the concentration of

(1) Substrate

(2) Product

(3) Enzyme

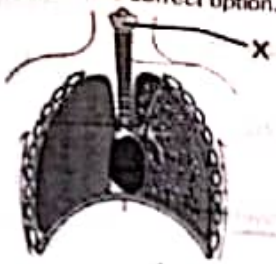
(4) Co-enzyme

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155. Identify the structure marked 'X' in the following figure and choose the correct option.



(1) Trachea
(2) Epiglottis
(3) Larynx
(4) Pharynx

156. The primary structure of a A, if imagined as a line, the left end is represented by the B amino acid (also called C amino acid) and the last amino acid is called the D amino acid.

Choose the option which correctly replace A, B, C and D

	A	B	C	D
(1)	Carbohydrate	Reducing end	Non-reducing end	C-terminal
(2)	Lipid	Non-reducing end	Reducing end	N-terminal
(3)	Protein	First	N-terminal	C-terminal
(4)	Protein	First	C-terminal	N-terminal

(1) (1)
(2) (2)
(3) (3)
(4) (4)

157. The factor which does not affect the catalytic activity of an enzyme is

(1) Temperature above the optimum temperature
(2) Temperature below the optimum temperature
(3) Concentration of its substrate
(4) Activation energy

158. During resting conditions, lung ventilation movements are primarily due to

(1) Costal muscles and diaphragm
(2) Costal muscles only
(3) Diaphragm only
(4) Abdominal muscles only

159. Choose the incorrect statement w.r.t. mechanism of breathing.

(1) Inspiration in mammals is an active process.
(2) Inspiration can occur, if atmospheric pressure is lower than intra-pulmonary pressure.
(3) Expiration can occur, if intra-pulmonary pressure is higher than the atmospheric pressure.
(4) Inspiration is initiated by the contraction of diaphragm.

160. During normal inspiration in humans, the

(1) Volume of pulmonary cavity increases
(2) Intra-pulmonary pressure becomes more positive than the normal
(3) Internal intercostal muscles contract which displaces the ribs from their original position in the antero-posterior axis
(4) Diaphragm becomes dome-shaped

161. The conducting part of human respiratory system does not facilitate

(1) Humidification of inhaled air
(2) Temperature adjustment of inhaled air
(3) Clearing of foreign particles from the inhaled air
(4) Exchange of gases between alveolar air and blood

162. Which of the following is not a lipid ?

(1) Butyric acid
(2) Palmitic acid
(3) Lecithin
(4) Arachidonic acid

163. Under normal physiological conditions the volume (mL) of air a healthy man can inspire or expire per minute is approximately

(1) 2000 – 2500 mL
(2) 6000 – 8000 mL
(3) 4000 – 4500 mL
(4) 3000 – 3500 mL

164. Arrange the following lung volumes and capacities (from lower to higher) in an adult man, under normal physiological conditions.

A. Tidal volume
B. Residual volume
C. Inspiratory reserve volume
D. Vital capacity

Choose the correct option.

(1) A < C < B < D
(2) A < D < B < C
(3) A < D < C < B
(4) A < B < C < D

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165. All of the following respiratory structures have capillary beds to facilitate gas exchange, **except**

(1) Moist cuticle of earthworm

(2) Skin of frogs

(3) Gills of fishes

(4) Tracheal system of cockroaches

166. The two strands of DNA are held together majorly by which bond?

(1) Coordination bond

(2) Glycosidic bond

(3) Hydrogen bond

(4) Peptide bond

167. Adult human haemoglobin consists of subunits of

(1) Two α and two β type

(2) One α and one β type

(3) Two α and one β type

(4) Three α and two β type

168. The total number of nitrogen atoms present in the ring of uracil is equal to the number of H-bonds present between

(1) Guanine and cytosine in DNA

(2) Adenine and thymine in DNA

(3) Trihydroxypropane and choline

(4) Monomeric units of arachidonic acid

169. Co-enzymes are

(1) Inorganic compounds only

(2) Tightly bound to the apoenzyme

(3) Associated with apoenzyme, usually occurring during the course of catalysis so as to form holoenzyme

(4) The proteinaceous part of holoenzyme

170. **Assertion (A):** Different groups of animals respire through different respiratory structures like gills, lungs, tracheal tubes, *etc.*
Reason (R): Mechanism of breathing vary among different groups of animals depending only on their habitats.
In the light of above statements, select the correct option

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are false

(3) (A) is true but (R) is false

(4) Both (A) and (R) are true but (R) is not the correct explanation of (A)

171. Thoracic chamber is ventrally, dorsally and laterally formed respectively by

(1) Vertebral column, ribs, and sternum

(2) Ribs, sternum, and diaphragm

(3) Sternum, vertebral column, and ribs

(4) Diaphragm, sternum, and ribs

172. Pulmonary ventilation takes place between

(1) Atmospheric air and alveolar air

(2) Lungs and pulmonary capillaries

(3) Tissue capillaries and tissue

(4) Tissue and cells

173. Part of which among the following is a common passage for food and air?

(1) Larynx

(2) Trachea

(3) Glottis

(4) Pharynx

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174. If we increase the temperature by 10°C (a change in temperature within the range of optimum value), the rate of an enzyme-catalysed reaction

(1) Will be doubled

(2) Will be tripled

(3) Remains unaffected

(4) Will be four times upto optimum temperature

175. On hydrolysis, a nucleoside would not yield

(1) Purines

(2) Pyrimidine

(3) Pentose sugar

(4) Phosphoric acid

176. Insulin is biochemically most similar to

(1) Chitin

(2) Trypsin

(3) Cellulose

(4) Lecithin

177. Read the given statements in context of humans.
(a) The lungs are situated in the thoracic cavity which is an anatomically air tight chamber.
(b) The anatomical setup of lungs in thorax is such that any change in the volume of the thoracic cavity will be reflected in the lung cavity.

(1) Only statement (a) is correct

(2) Only statement (b) is correct

(3) Both statements (a) and (b) are correct

(4) Both statements (a) and (b) are incorrect

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178. α-amino acids are substituted

(1) Glucose

(2) Methane

(3) Alcohol

(4) Formic acid

179. The respiratory volume/capacity of air that **cannot** be measured by using a simple spirometer is

(1) Inspiratory capacity

(2) Tidal volume

(3) Expiratory reserve volume

(4) Functional residual capacity

180. The structure responsible for the production of sound is

(1) Pharynx

(2) Larynx

(3) Trachea

(4) Bronchi

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